



Polymer Processing 2

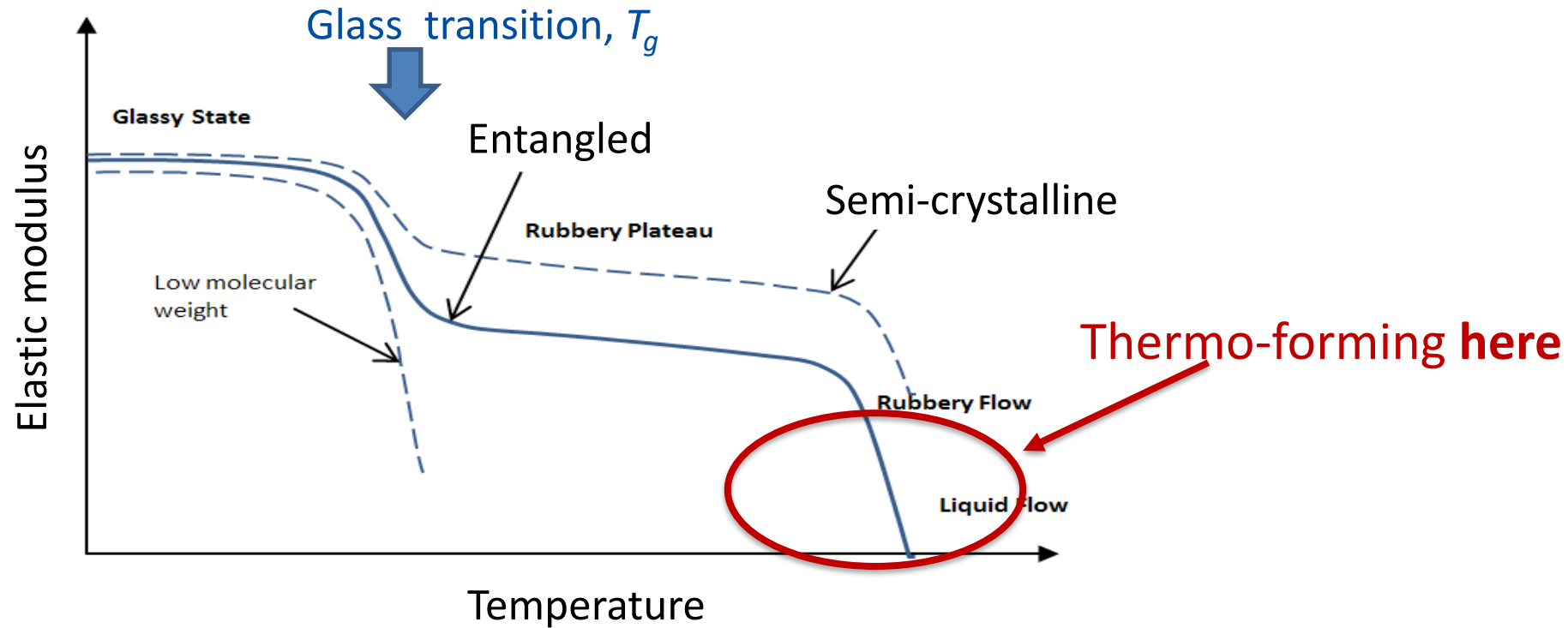
Presented by Dr Adijat 'Wumi Inyang

Polymer processing methods that use heat

- Melting or softening of polymers
(thermoplastics & elastomers)
- Thermo-forming (moulding and extrusion processes)
- Films, fibres and foams
- Additive manufacturing

Mechanical behaviour

Thermo-forming when polymer can plastically deform easily



- Large temperature range between T_g and T_m useful for thermo-forming
- Will depend on the molecular configuration, Mw and ability to crystallize

Thermo-forming

Step 1

- Preform (polymer and additive) placed in mould cavity



Step 2

- Mould heated and pressure applied

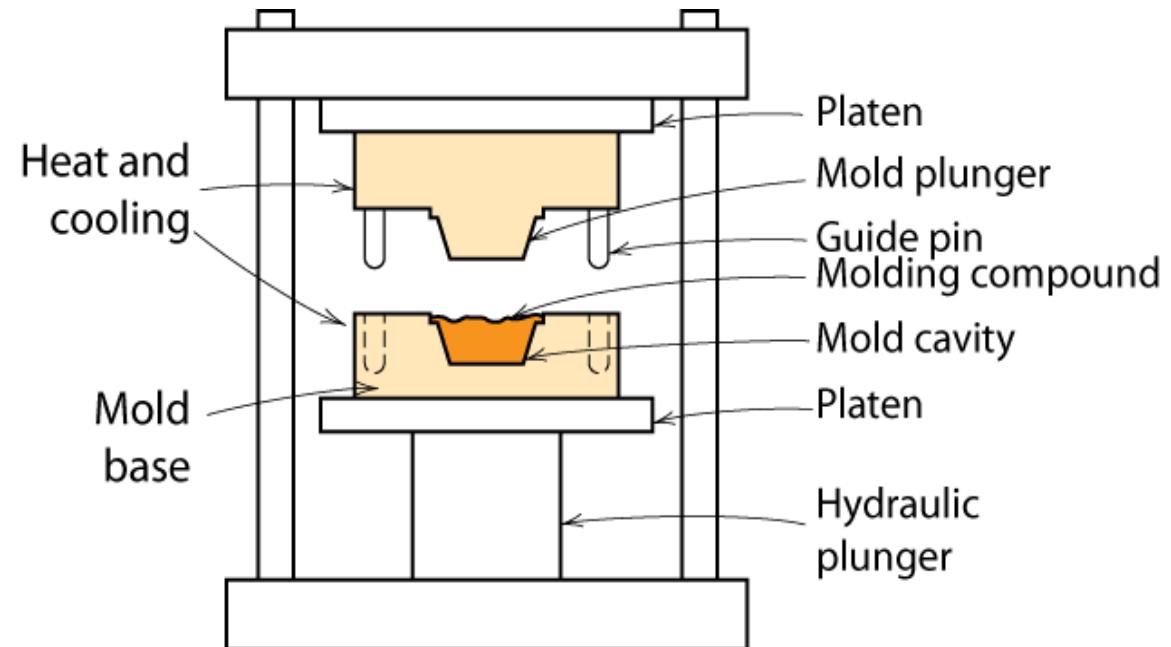


Step 3

- Polymer deforms to take shape of mould

Compression moulding

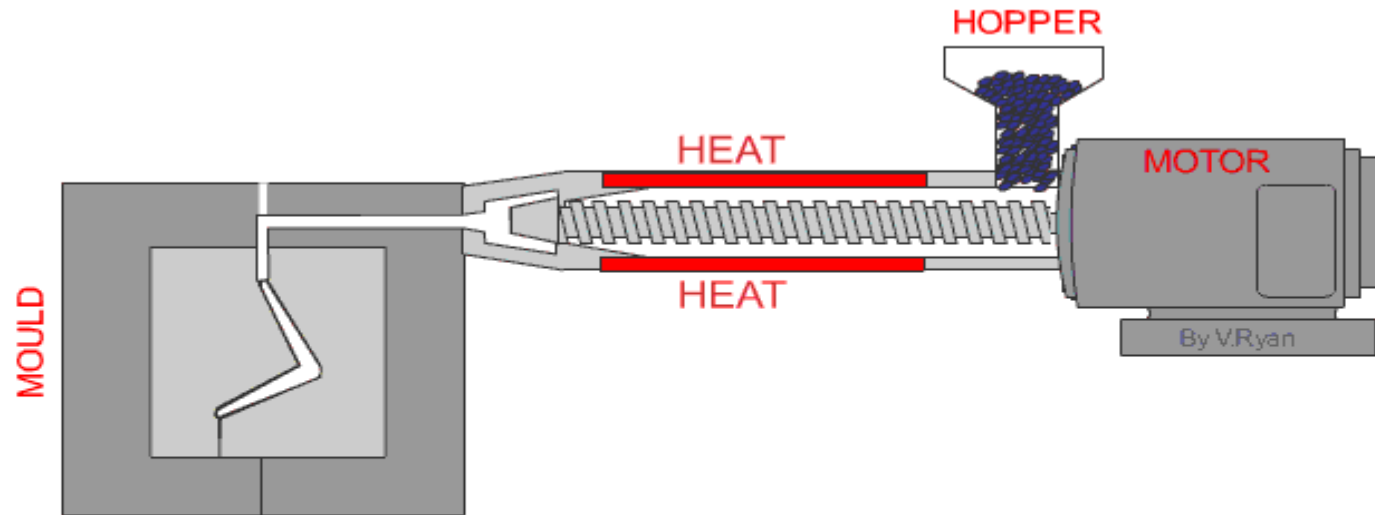
- Starting from preform (e.g. sheet material)
- Polymer heated above T_g not above T_m
- Cooling relatively slow
- Cycle time = relatively slow ($>10s$)



Thermo-forming

Injection moulding

- Polymer heated above T_m
- Cooling relatively fast
- Cycle time = fast ($< 10s$)

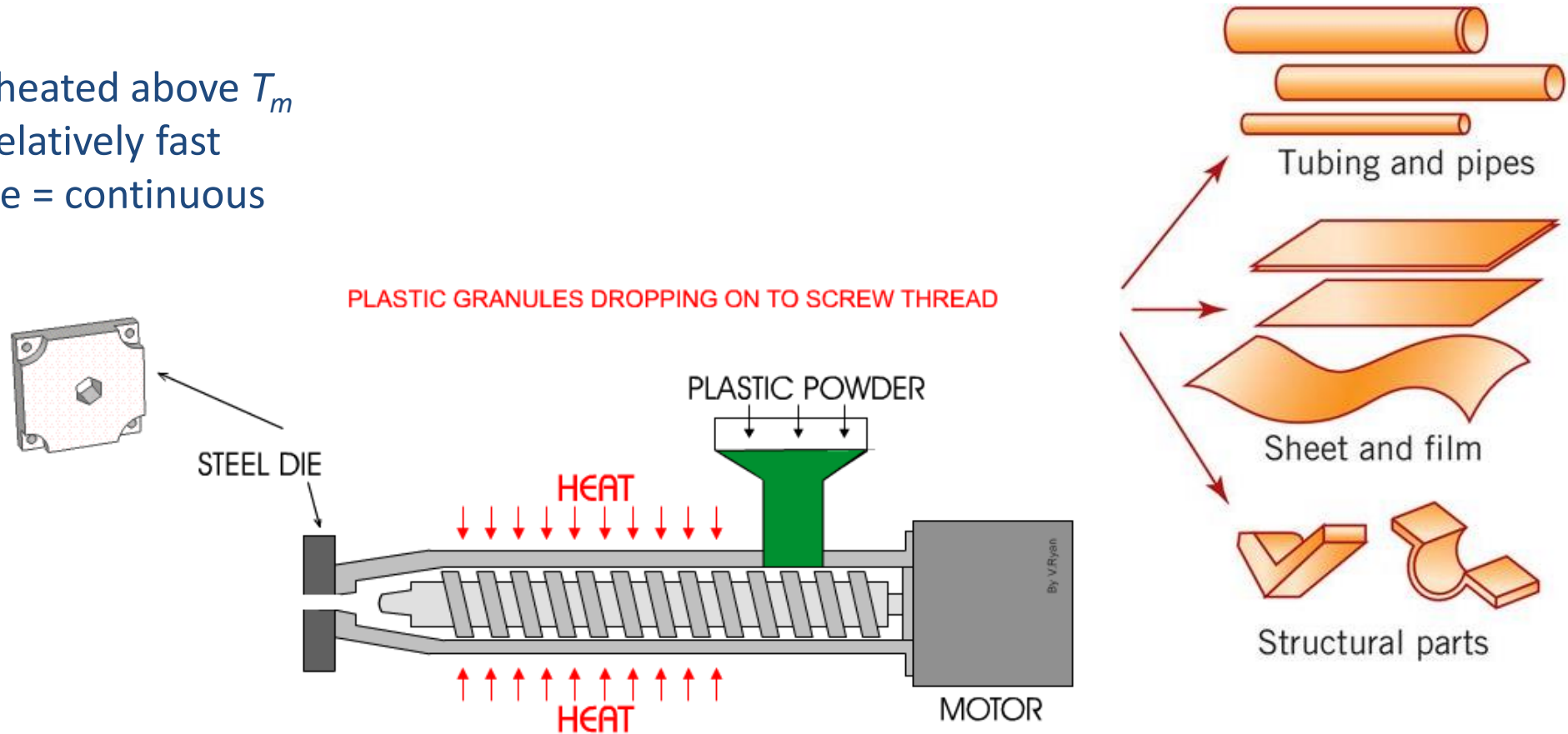


- Ram retracts, plastic pellets drop from hopper into barrel
- Ram forces plastic into the heating chamber where the **plastic melts** as it moves forward
- Molten plastic is forced under pressure (injected) into the mold cavity where it takes up the shape of the mold

Thermo-forming

Extrusion

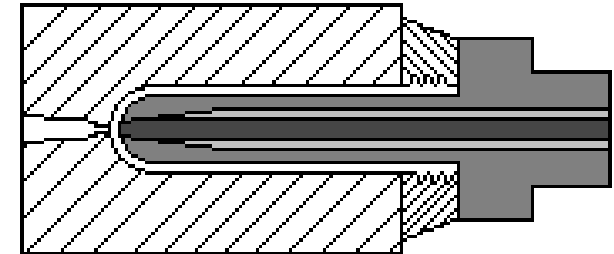
- Polymer heated above T_m
- Cooling relatively fast
- Cycle time = continuous



- Plastic pellets drop from hopper onto the turning screw
- Plastic pellets melt as the turning screw pushes them forward by the heaters
- Molten polymer is forced under pressure through the shaping die to form the final product (extrudate) 6

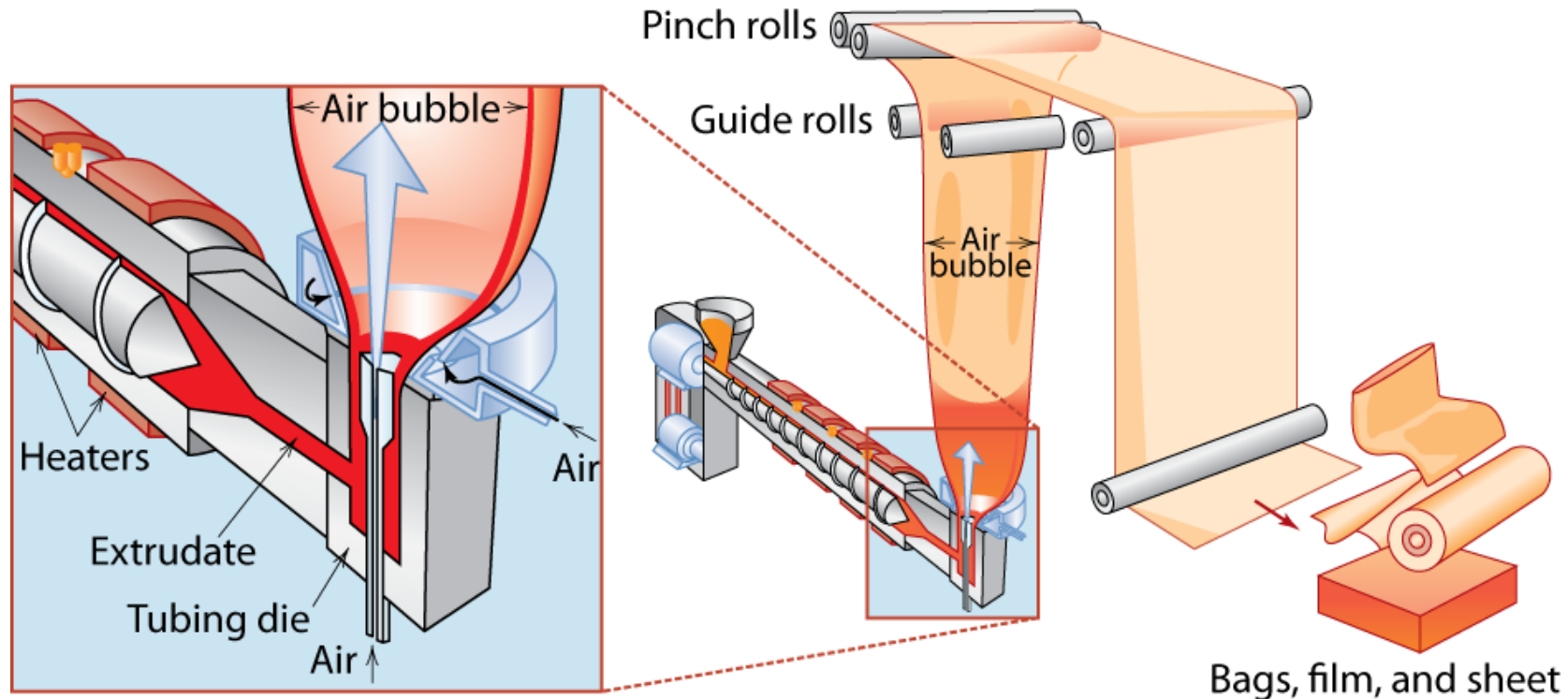
Blow moulding

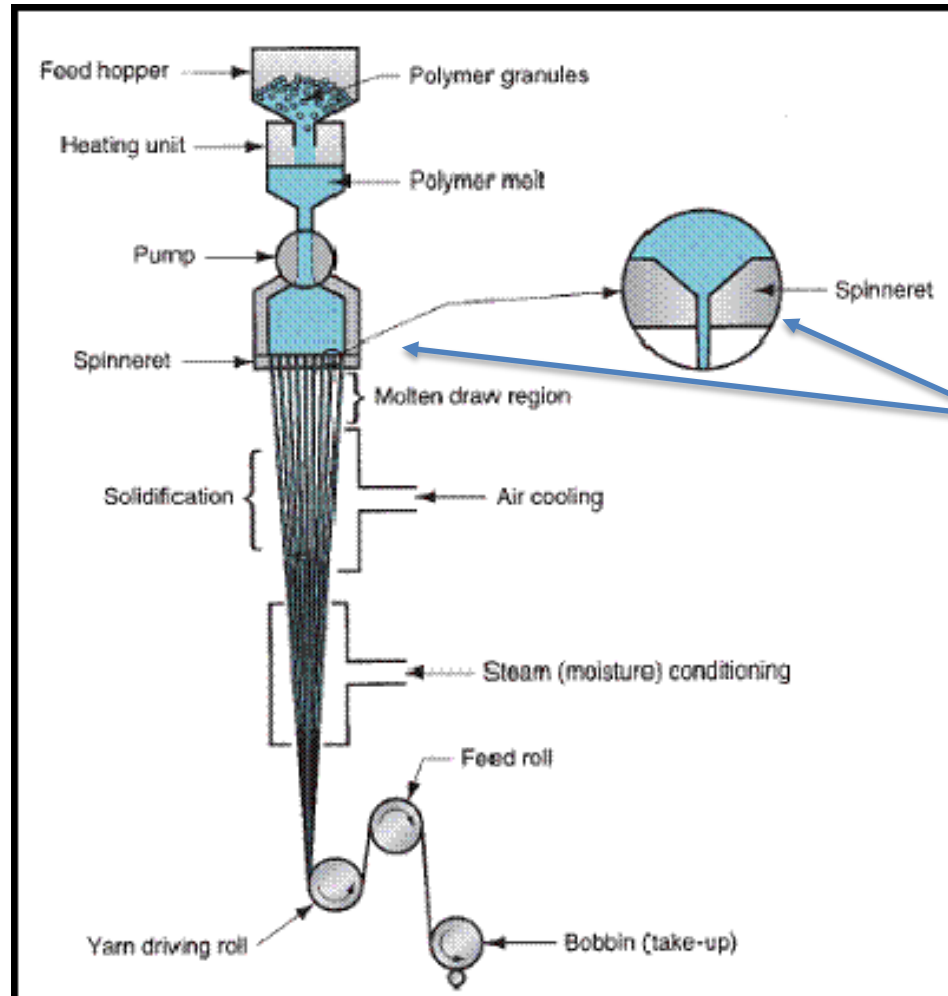
- Preform (**parison**) is injection moulded or extruded
 - Polymer heated above T_g not above T_m
 - Orientation of crystallites
 - Cycle time = relatively fast ($\sim 10s$)
-
- Molecules are aligned mechanically by the flow induced during the shaping process
 - All plastic bottles are made this way



Blow moulding – films and bags

- Tube is extruded
- Polymer stretched above T_g not above T_m
- Cycle time = continuous



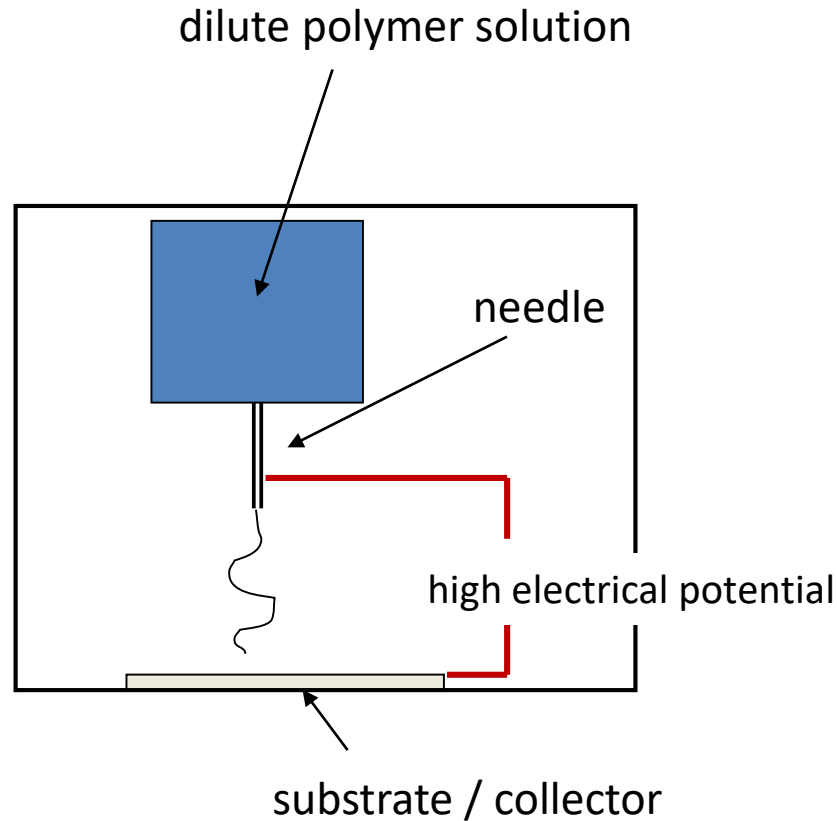


Fibre spinning

- Polymer heated above T_m
- Cooling relatively fast
- Cycle time = continuous
- Fibre drawn as it cools



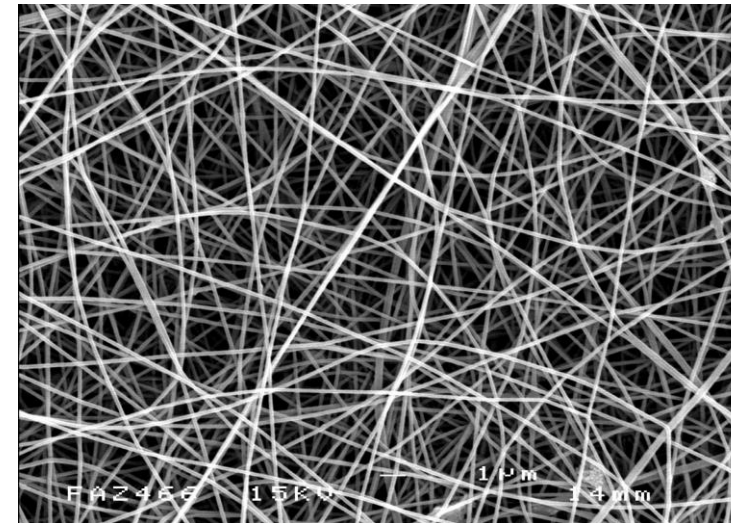
Spinneret



The electro-spun nano-fibres have unusual structure and surface properties

Electro-spinning

- Polymer in solvent
- Electrical potential forms the liquid into a very thin stream
- Solvent evaporates
- Fibre drawn as it cools
- Cycle time = continuous



Electro-spun nanofibres
10 μ m

Foam-forming

Foamed polymers

- 'Blowing' agent mixed with polymer
- On heating the blowing agent decomposes releasing gas (H_2)
- Gas bubbles formed in molten or viscous liquid polymer
- Cools to create a foam

Thermoplastics and thermosets and elastomers can be formed into foams

- Polyurethane
- Polystyrene
- Polyvinyl chloride
- Rubber



Additive manufacturing

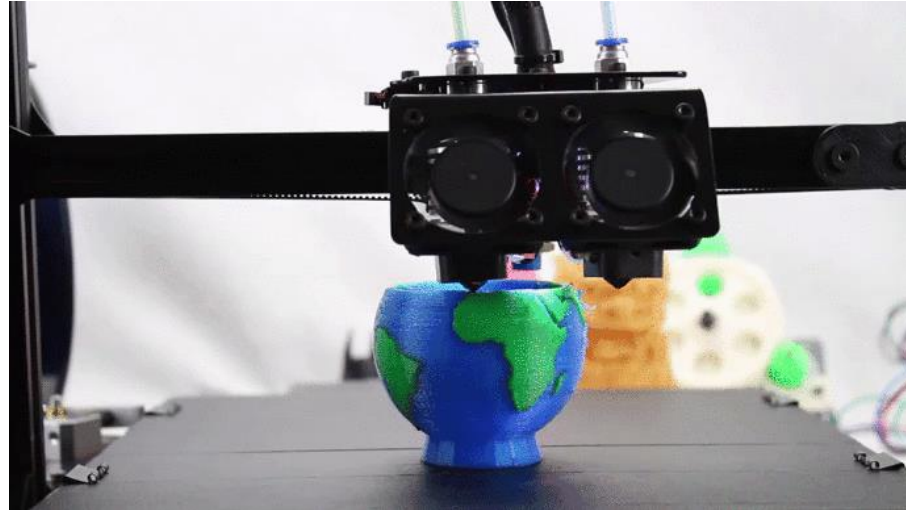
3D printing

Thermoplastics and thermosets and elastomers can be 3D printed
There are now many different methods:

- **Fused Deposition Modelling (FDM)**

Continuous filament of polymer is melted and fused by moving printer

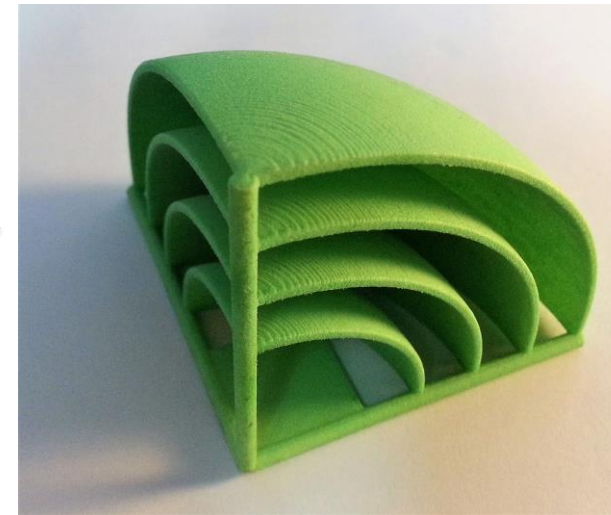
– thermoplastics e.g. ABS, PLA, PET



- **Selective Laser Sintering (SLS)**

Layers of polymer powder is fused together by a moving laser

– thermoplastics e.g. PA



- **Stereo lithography (SLA)**

Liquid polymer is cross-linked by UV light

– thermosets e.g. cyanoacrylates

Environmental issues

Good - polymers have many advantages over metals and ceramics 😊

- **Versatile** materials with wide range of properties
- Low processing temperatures and short process times
= **Low embedded energy**
- **Re-use** of materials possible

Bad - polymer pollution 😞

Many polymers do not break down or degrade in the environment and are **single use** only

Solutions:

- Controls on use and capture of waste plastic
- Use of bio de-gradable or reusable materials



Polymer processing:

There are many ways to process polymer materials

Their properties change above and below T_g and with the degree of crystallinity

The processing will change their structure and properties depending on the temperature and time to form them

Learning Outcomes



What can you do now?

- Select polymers based on their chemical structure and properties for different applications
- Design polymer materials and products
- Understand how polymers change their behaviour with temperature or strain rate
- Understand how to process polymers
- Select processing methods and heat treatments to achieve the desired performance of polymers